Solutions(vector)

1. (a) $R^2 = A^2 + B^2 + 2AB\cos\theta$

   Substituting, $A = (x+y)$, $B = (x-y)$ and $R = \sqrt{(x^2+y^2)}$

   we get $\theta = \cos^{-1}\left(-\frac{(x^2+y^2)}{2(x^2-y^2)}\right)$

2. (d) $\Delta a = 2a\sin\left(\frac{\theta}{2}\right) = 2a \times \sin 45° = \sqrt{2}a = \sqrt{2}\frac{v^2}{R}$

3. (b) $\Delta v = 2v\sin\left(\frac{\theta}{2}\right) = 2v\sin 20°$

4. (c) $\hat{n} = \frac{\vec{A}\times\vec{B}}{|\vec{A}\times\vec{B}|} = \frac{8\hat{i}-8\hat{j}-8\hat{k}}{8\sqrt{3}} = \frac{1}{\sqrt{3}}(\hat{i}-\hat{j}-\hat{k})$

   There are two unit vectors perpendicular to both $\vec{A}$ and $\vec{B}$ they are $\hat{n} = \pm\frac{1}{\sqrt{3}}(\hat{i}-\hat{j}-\hat{k})$

5. (d) $AB\cos\theta = AB\sin\theta \Rightarrow \tan\theta = 1 \therefore \theta = 45°$

   $\therefore |\overline{C}| = \sqrt{A^2+B^2+2AB\cos 45°} = \sqrt{A^2+B^2+\sqrt{2}AB}$

6. (c) Vector perpendicular to *A* and *B*, $\vec{A}\times\vec{B} = AB\sin\theta\,\hat{n}$

   $\therefore$ Unit vector perpendicular to *A* and *B*

   $\hat{n} = \frac{\vec{A}\times\vec{B}}{|\vec{A}|\times|\vec{B}|\sin\theta}$

7. (b) $R_{net} = R + \sqrt{R^2+R^2} = R + \sqrt{2}R = R(\sqrt{2}+1)$
8. (d)
9. (d) $\Delta v = 2v\sin\left(\frac{90°}{2}\right) = 2v\sin 45° = 2v\times\frac{1}{\sqrt{2}} = \sqrt{2}v$

   $= \sqrt{2}\times r\omega = \sqrt{2}\times 1\times\frac{2\pi}{60} = \frac{\sqrt{2}\pi}{`30}\ cm/s$

10. (d) If a vector quantity has zero magnitude then it is called a null vector. That quantity may have some direction even if its magnitude